Analytical Problem Solving

Activity

(1). Given:

$G(x) = 24$

$G(y) = 18$

To find: 1. Gradient magnitude
2. Gradient direction.

1. Gradient magnitude

$$\boxed{G = \sqrt{G_x^2 + G_y^2}}$$

$$= \sqrt{(24)^2 + (18)^2}$$

$$= \sqrt{576 + 324}$$

$$= \sqrt{900}$$

$$\boxed{G = 30}$$

2. Gradient direction

$$\boxed{\theta = \tan^{-1}\left(\frac{G_y}{G_x}\right)}$$

$$\theta = \tan^{-1}\left(\frac{18}{24}\right)$$

$$\theta = \tan^{-1}(0.75)$$

$$\boxed{\theta = 36.870}$$

② Non - Maximum Suppression:

pixel :	P_1	P_2	P_3	P_4	P_5
Magnitude	25	56	90	60	30

$P_1 = 25 < P_2 \rightarrow 0$

$P_2 = 50 < P_3 \rightarrow 0$

$P_3 = 90 > P_2 , 4$

$P_4 \rightarrow$ keep 90

$P_4 = 60 < P_3 \longrightarrow 0$

$P_5 = 30 < P_4 \rightarrow 0$

NMS

pixel	P_1	P_2	P_3	P_4	P_5
NMS	0	0	90	0	0

$P_3 = 90$ is retained.

③ Given :- High Threshold = 100

Low Threshold = 50

The gradient magnitude are.

Pixel	P_1	P_2	P_3	P_4	P_5	P_6.
magnitude :	135	90	45	20	70	20

$M \geq 100 \longrightarrow$ strong

$50 \leq M < 100 \longrightarrow$ weak.

$M < 50 \longrightarrow$ Non - edge.

Pixel	Magnitude	classification.
P_1	135	strong
P_2	90	weak

P_3	45	non-edge.
P_4	120	strong
P_5	70	weak
P_6	20.	non-edge.

4) Given :-

Pixel	P_1	P_2	P_3	P_4	P_5	P_6	P_7
Magnitude	130	95	40	115.	65	25	105

High Threshold = 100
low Threshold = 50

Strong edge :- $m \geq 100$

$P_1 = 130$, $P_4 = 45$, $P_7 = 105$

weak edge :- $50 \leq m \leq 100$

$P_2 = 95$, $P_5 = 65$

Non edge :- $M < 50$

$P_3 = 40$, $P_6 = 25$

Hysteresis :-

$P_2 \rightarrow P_1$ (retained)

$P_5 \neq$ edge - suppressed

$\therefore$ Final edge map is P_1, P_2, P_4, P_7

5) Hough Circle Transform :-

$$r = \sqrt{(x-a)^2 + (y-b)^2}$$

(i) edge points :-

$P_1 = (5, 4)$

$P_2 = (7,6)$

$C_1 = (2,4)$

$C_2 = (4,3)$

For $C_1 = (2,4)$

P_1 and C_1 : $r = \sqrt{(5-2)^2 + (4-4)^2}$

$= \sqrt{3^2 + 0^2}$

$= \sqrt{9}$

$\boxed{r = 3}$

P_2 and C_2 : $r = \sqrt{(7-2)^2 + (6-4)^2}$

$= \sqrt{25+4}$

$= \sqrt{29}$

$\boxed{r = 5.39}$

P_1 and C_2 : $r = \sqrt{(5-4)^2 + (4-3)^2}$

$= \sqrt{1+1}$

$= \sqrt{2}$

$\boxed{r = 1.41}$

P_2 and C_2 : $r = \sqrt{(7-4)^2 + (6-3)^2}$

$= \sqrt{9+9}$

$= \sqrt{18}$

$\boxed{r = 4.24}$

Calaulate the radins:

Given:- $r = (8,6)$

$c = (4,3)$

$r = \sqrt{(x-a)^2 + (y-b)^2}$

$= \sqrt{(8-4)^2 + (6-3)^2} = \sqrt{25}$ $\boxed{r = 5}$

Calculate the radius:

$C = (5,4)$

$P = (8,8)$

$r = \sqrt{(8-5)^2 + (8-4)^2}$

$= \sqrt{3^2 + 4^2}$

$= \sqrt{9+16}$

$= \sqrt{25}$

$\boxed{r = 5}$